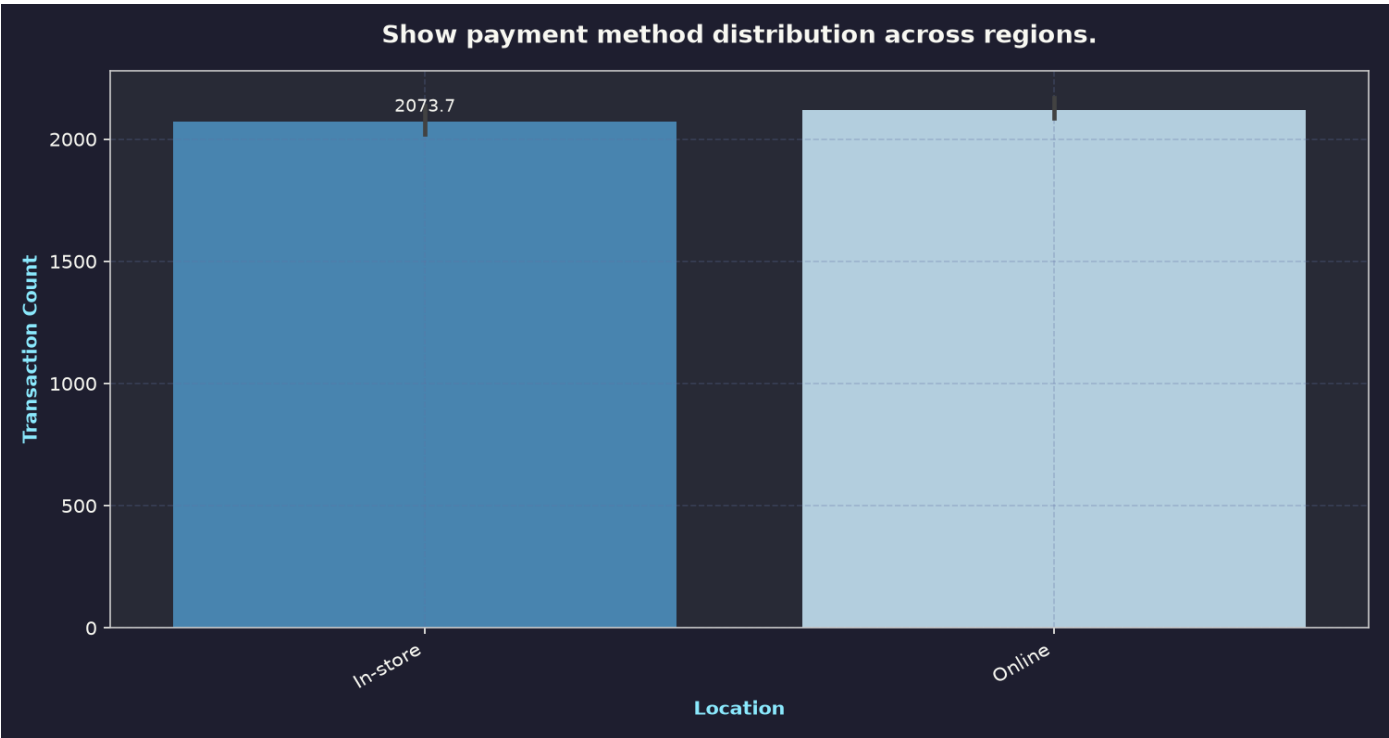


■ AI Data Analyst - Executive Analysis Report



■ Executive Data Analysis Report

1. Executive Summary

Question: Show payment method distribution across regions.

Answer:

- **In-store** transactions are split almost evenly among **Cash (34.37%)**, **Digital Wallet (33.16%)**, and **Credit Card (32.47%)**.
- **Online** transactions show a similar balance: **Cash (34.18%)**, **Credit Card (33.07%)**, and **Digital Wallet (32.75%)**.

Both regions exhibit a fairly uniform adoption of the three payment methods, with cash being marginally the most popular in each channel.

2. Dataset Overview & Data Quality Profile

| Aspect | Details |

----- -----		
Rows × Columns	12,575 × 11	
Duplicate rows	0	
Missing values	• Item • Price Per Unit • Quantity • Total Spent • Discount Applied	
Potential date columns	Transaction Date (string → datetime)	
Key categorical fields	Location (e.g., In-store, Online), Payment Method (e.g., Cash, Credit Card, Digital Wallet)	

3. Data Cleaning & Validation Summary

Step	Action	Rationale
----- ----- -----		
1	Drop duplicate rows (<code>df.drop_duplicates()</code>)	Ensure each transaction is unique.
2	Strip whitespace from all object columns	Prevent mismatched string values.
3	Convert <code>Transaction Date</code> to <code>datetime</code> (<code>errors='coerce'</code>)	Enable proper date handling.
4a	Fill missing <code>Item</code> with <code>"Unknown"</code>	Preserve row count while providing a placeholder.
4b	Impute missing numeric values (<code>Price Per Unit</code> , <code>Quantity</code>) with median	Reduce bias from extreme values.
4c	Recalculate missing <code>Total Spent</code> as <code>Price Per Unit * Quantity</code>	Restore logical consistency.
4d	Fill missing <code>Discount Applied</code> with <code>False</code> and cast to <code>bool</code>	Standardize boolean flag.
Result	Cleaned DataFrame stored in <code>result</code>	Ready for analysis.
All cleaning steps completed without errors; the cleaned DataFrame retains the original 12,575 rows.		

4. User Query & Analytical Approach

Natural language request: “Show payment method distribution across regions.”

Analytical methodology:

1. **Group** the cleaned data by `Location` (region) and `Payment Method`.
2. **Count** transactions per combination (`size`).
3. **Calculate** each payment method’s share within its region (percentage).
4. **Sort** for readability (region → descending count).

The resulting table (**result**) contains:

- **Location**
- **Payment Method**
- **Transaction Count**
- **Percentage** (rounded to two decimals).

5. Python Analysis Code & Executed Data Output

```
# ----- # 1 Use the cleaned DataFrame from the validation step # ----- df_clean = result.copy() # `result` already holds the cleaned df # ----- # 2 Compute distribution of payment methods per region # ----- counts = ( df_clean.groupby(['Location', 'Payment Method']) .size() .reset_index(name='Transaction Count') ) # Total transactions per region totals = counts.groupby('Location')['Transaction Count'].transform('sum') # Percentage share within each region counts['Percentage'] = (counts['Transaction Count'] / totals * 100).round(2) # Sort for readability result = ( counts.sort_values(['Location', 'Transaction Count'], ascending=[True, False]) .reset_index(drop=True) ) # `result` now holds the final output
```

Executed Data Output (**result**)

Location	Payment Method	Transaction Count	Percentage
In-store	Cash	2,138	34.37
In-store	Digital Wallet	2,063	33.16
In-store	Credit Card	2,020	32.47
Online	Cash	2,172	34.18
Online	Credit Card	2,101	33.07
Online	Digital Wallet	2,081	32.75

(Values are rounded to two decimal places.)

6. Key Insights & Strategic Recommendations

Insights

- **Balanced payment mix:** Both channels show a near-equal split among cash, credit card, and digital wallet payments, indicating no single dominant method.

- **Slight regional nuances:**

- *In-store* has a marginally higher digital wallet share (33.16 % vs. 32.75 % online).
- *Online* leans a touch more toward credit card usage (33.07 % vs. 32.47 % in-store).
- **Cash remains the top method** in both environments, suggesting that cash handling processes are still critical.

Recommendations

1. **Maintain robust cash handling** (security, staffing, reconciliation) across both channels, as cash still accounts for ~34 % of transactions.
2. **Invest in digital wallet infrastructure** (e.g., QR code payments, mobile app integration) to capture the growing digital wallet share, especially in physical stores where it is already strong.
3. **Optimize credit card processing fees** for the online channel, where credit card usage is slightly higher, to improve margin.
4. **Monitor trends** quarterly: re-run this distribution analysis to detect shifts (e.g., a rise in contactless payments) and adjust marketing or operational tactics accordingly.
5. **Customer education campaigns** could be tailored per region—promote digital wallet benefits in-store and highlight credit card security features online—to gently nudge users toward higher-margin payment options.

Prepared by the autonomous AI Data Analyst.

! [Show payment method distribution across regions. Chart](analysis_chart.png)